The Journal of Open Source Software

Cobwood: Enhancing Forest Economics Model Reusability Through labelled Panel Data Structures

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DOI: 10.xxxxxx/draft

Software

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Editor: Ethan White

Submitted: 25 June 2025
Published: unpublished

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Summary

Managing forest ecosystems effectively requires long-term foresight into global wood markets. This planning relies on macroeconomic forest sector models spanning multiple countries over extended time periods. The cobwood package introduces a panel data structure based on labelled N-dimensional arrays from the Xarray package, including output storage to NetCDF files. The comprehensive metadata for country, product, time coordinates along with units enhances source code clarity and facilitates model inspection. To demonstrate cobwood's practical application, we present a reimplementation of the Global Forest Products Model (GFPMx). The reusable data structure positions cobwood as an ideal component for integration into a greater modelling tool chain.

Statement of need

Trees grow over decades or centuries and wood markets may be very localized. Yet markets for processed wood and paper products are interconnected at the global scale requiring, decision makers to understand long-term forecasts of global wood consumption, production and trade. This need has led forest economists to develop macroeconomic models of the forest sector. Several global forest sector models currently exist, including the Global Forest Products Model (GFPM)(Buongiorno et al., 2003), the European Forest Institute Global Trade Model (EFI-GTM)(Kallio et al., 2004), the Global Forest and Agriculture Model (G4M)(Gusti, 2020), the Global Forest Trade Model (GFTM) (Jonsson et al., 2015) and an adaptation called Timba (TI-FSM et al., 2025). There are also multiple regional and national forest sector models.

These macroeconomic models organize market datasets as panel data with country and time dimensions, containing information on production, consumption, and trade for products such as roundwood, sawnwood, wood panels, pulp, and paper. However, current modeling software often lacks proper panel data structures, instead using partial labeling approaches with unclear variable names that make source code difficult to interpret. Many models are not open source, and limited data labeling makes model outputs difficult to reuse.

Adjacent research fields including forest management, vegetation dynamics, and life cycle analysis need estimates of future roundwood harvest and wood products consumption. Model transparency helps these communities determine whether existing models are suitable for specific policy questions or can be modified to simulate new drivers influencing forest products markets.

Input, output

A yaml file in the `cobwood_data/scenario` directory defines the particular input data used for a given scenario. Cobwood can load input data from any tabular source that pandas support.

For instance, the GFPMx data is stored inside a single Excel spreadsheet containing many sheets for consumption, production, import, export, and prices of major forest products. A script first converts sheets to CSV files, which the `GFPMXData.convert_sheets_to_dataset` then transforms into an Xarray data structure. Other methods make it possible to load forest products market data from the FAOSTAT API and to transform them into xarray datasets.

The `write_datasets_to_netcdf` combines many products 2D datasets into one larger 3D dataset, by adding a third coordinate called "`product`" before saving the model output datasets to NetCDF files. These files include metadata labels for units. While not commonly used in economics, the NetCDF format is standard in earth systems modelling, making it ideal for integrated modelling systems.

Data structure and implementation

Figure 1 illustrates the data structure:

- Global consumption, production, trade flows, and prices for all countries, all years and for each forest product are stored as an Xarray dataset (e.g., `model["sawn"]` for sawnwood)
- Within each dataset for one product, specific variables are accessible as two-dimensional arrays with country and year coordinates (e.g., `model["sawn"]["cons"]` for consumption)

To explore available variables, users can access the `variables` property (e.g., `model["sawn"].variables`). Array properties are used to store metadata, the example below displays the roundwood production unit :

```
model["indround"]["prod"].unit
# '1000m3'
```

The cobwood model has been used to produce scenario analysis Mubareka et al. (2025) and Rougieux et al. (2024). The first model programmed inside cobwood is a reimplementation of a simple forest sector model called GFPMx (Buongiorno, 2021). Labelled data arrays allow developers to write Python functions that closely mirror the mathematical equations found in the academic papers describing the models, with explicit time and country dimensions. For example the demand function in `cobwood/gfpmx_equations.py` is implemented on an `xarray` dataset `ds` where a dependent variable such as GDP is selected for all countries at time `t` with `ds["gdp"].loc[ds.c, t]`.

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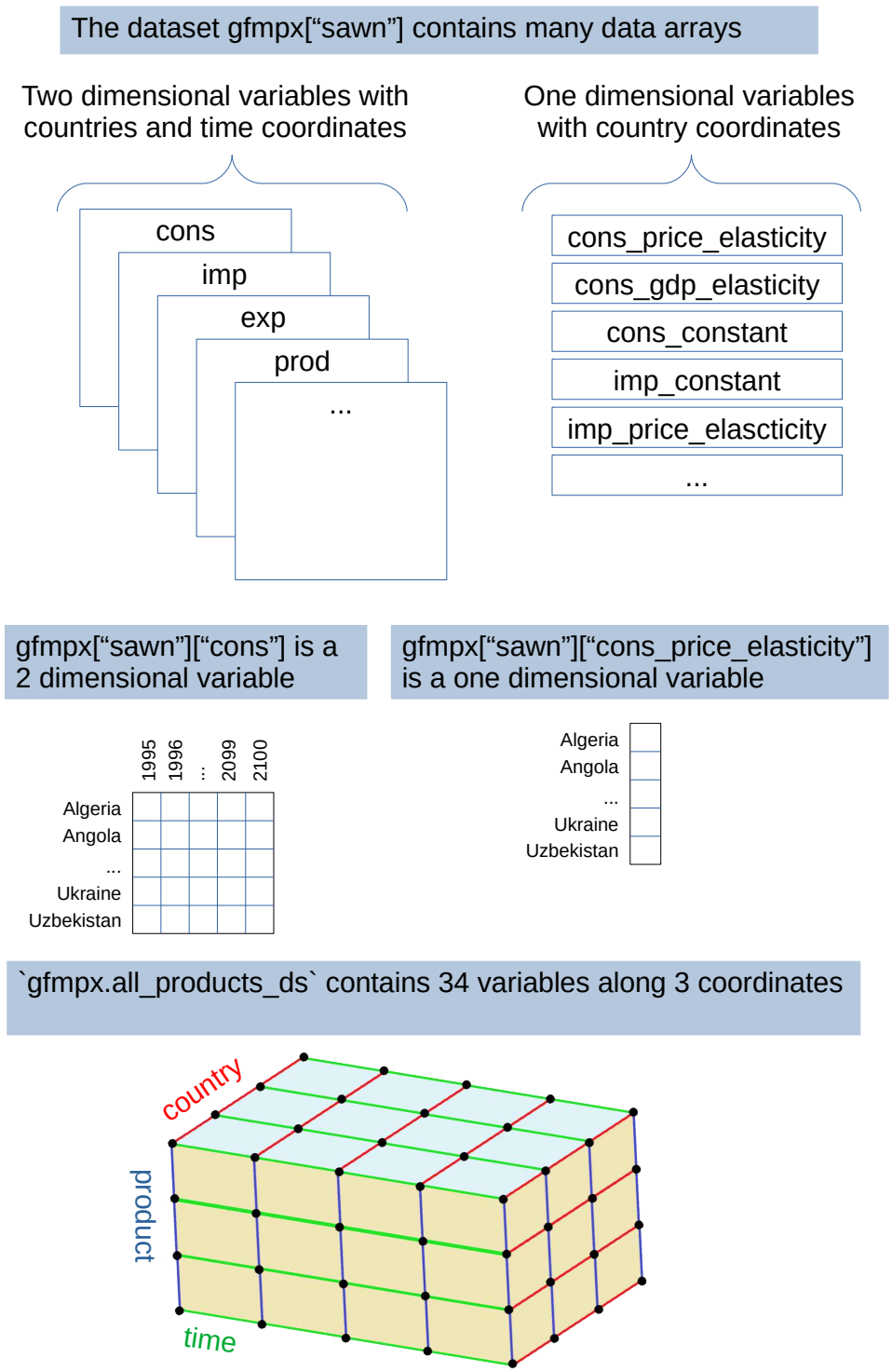

Figure 1: Data structure

Model run

The following code instantiates a GFPMX model object from a scenario yaml file. The `rerun=True` argument erases previous model runs, while `compare=True` compares output with the reference Excel implementation of GFPMx

```
from cobwood.gfpmx import GFPMX
gfpmxb2021 = GFPMX(scenario="base_2021", rerun=True)
gfpmxb2021.run(compare=True, strict=False)
```

The model output data is saved inside the model's `output_dir` directory. When re-using the model later, specify the argument `rerun="False"` (default) to load the output data without the need to run the model.

Visualisation

The following python code draws a faceted plot of industrial roundwood consumption, import, export, production and price with one line by continent.

```
gfpmxb2021.facet_plot_by_var("indround")
```

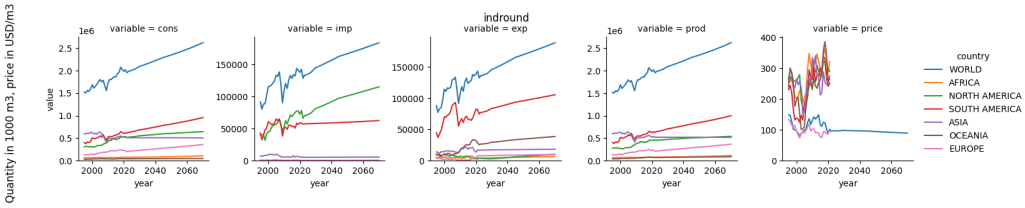

Figure 2: Industrial roundwood variables by continent

Specify the country argument to get one line by country

```
gfpmxb2021.facet_plot_by_var("indround", countries=["Canada", "France", "Japan"])
```

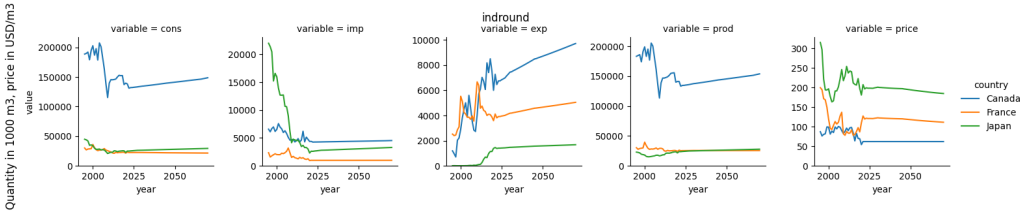

Figure 3: Industrial roundwood variables by country

Conclusion

The cobwood package represents macroeconomic forest products market data as N-dimensional labelled data arrays. The data structure incorporates comprehensive metadata and coordinates improving source code readability and model transparency. Additionally, the scenario configuration file enables comparison of different model implementations across variations of input parameters. Furthermore, model outputs are saved to NetCDF files, which preserve dimensions and metadata. This data structure will be reused to implement many other forest sector models. Ultimately, the goal is to facilitate the integration of forest sector models as components of interdisciplinary modelling tool chains.

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